

CONTROLLED ENVIRONMENT BERRY PRODUCTION INFORMATION

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Fertigation

Irrigation (fertigation) is a key management technique in high quality strawberry production. However, there is much room to optimize the nutrient solution and its application. The following is information based on our study at the University of Arizona and at the Ohio State University and may not be directly applicable to your conditions.

Nutrient recipe. There is limited information available for nutrient solution designed for strawberry soilless cultivation. We have used two different Japanese strawberry nutrient solutions and the concentrations of major ions are shown in the following table. We combine these major nutrients with a commonly used minor nutrients (Jensen, unpublished). Compared to those solutions widely used for tomato and other greenhouse crops (EC = ~2.4 dS/m at the full strength), both strawberry solutions have much lower overall concentration (EC = ~1.0 dS/m at the full strength). Adding a small percent of nitrogen in NH4 form helps to minimize the increase of pH in the root zone. However, too high NH4 rate should be avoided for production, as plants become too vegetative (in fact, we use increased nitrogen with NH4 form for runner propagation).

Table 1. Major essential nutrients (mg/L or ppm) for fruit production

	NO ₃ -N	NH ₄ -N	PO ₄ -P	K	Ca	Mg	SO ₄ -S
Yamazaki strawberry formula (full strength)	70*	7*	15	117	40	12	16
Tochigi strawberry formula (full strength)	111	10	30	156	86	22	11

*We modify NO₃-N and NH₄-N concentrations to 80 and 20 mg/L for propagation (runnering).

Table 2. Minor essential nutrients (mg/L or ppm)

	Fe (chelated)	B	Mn	Cu	Mo	Zn
All formula	2	0.34	0.55	0.05	0.05	0.33

Nutrient solution pH (5.5-6.0). The nutrient solution pH should be in the range (5.5 – 6.0). However, root zone pH is affected by the substrate pH, fertilizer cations/anions ratio, plant uptake and source water alkalinity. In some cases, nutrient solution pH needs to be lower than 5.5 to maintain the root-zone in optimum range. When root zone (or drainage solution)

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pH exceeds 7.0, young leaves turn yellow-green, a typical symptom of iron deficiency, which restricts overall growth of plants.

Actively growing roots in hydroponics take up hydrogen ions (H⁺) during NO₃-N uptake. This increases root-zone pH, to become more basic. However, when roots are old and not functioning, we see pH go down mainly due to oxidation and deterioration of roots. Daily measurements of drip and drain nutrient solutions are essential to keep the plants healthy and actively growing roots.

Nutrient solution EC (1.0 dS/m or less). EC is electrical conductivity and is an indicator of the total concentration of ions dissolved in the solution. Many experts of greenhouse soilless strawberry production suggest that strawberries are sensitive to salts accumulated in the root zone. When the root-zone EC exceeds 1.2 dS/m, the growers often flush the root-zone with water to wash off the excessively accumulated salts in the root zone. We are often asked whether keeping higher EC levels can improve the fruit flavor (just like we do for tomato crops). Although we notice fruit becomes slightly 'water-rich' flavor immediately after this wash-off practice, we do not recommend keeping EC higher than recommended as salt-stress could cause a prolonged negative effect (given the sensitivity of strawberry to salt stress).

Irrigation management in substrate-based system. A typical irrigation set up consists of concentrated fertilizer injectors, drip irrigation system with pressure compensated emitters, and lysimeter set-up to measure EC, pH, and volume of drip and drain nutrient solution. Strawberry plants are sensitive to 'too wet' or 'too dry' conditions. In order to avoid large fluctuation of the moisture content, we irrigate frequently with a very small amount (33 mL per plant at a time), rather than only one or two large irrigation events per day. The total irrigation applications per day depends on the total ET (evapotranspiration rate) as affected by the greenhouse environments (solar radiation, temperature, and VPD) as well as the plant leaf area index, but it has been between 6 to 12 irrigation events per day (~200 mL to ~400 mL per plant per day) in our winter/spring crop production.

Drainage percent (20-30%). Assuring a good drainage percent is a standard key practice in root zone management for soilless substrate culture. The recommended drainage percentage for strawberry is 20-30%, a similar level to tomato and other crops. We also try to not have any drainage in the early part of the day. Having a measurable amount of drainage at the first few irrigations of the day is an indication that the root zone is too wet.


Hydroponic strawb

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02:46

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